S.S.V.P.S's B. S. Deore Polytechnic, Dhule

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SEMINAR AND PROJECT INITIATION COURSE

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Topic : AI Based Rainwater Harvesting Potential Predector

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Abstract

- Water scarcity and irregular rainfall have increased the need for effective water conservation. Rainwater harvesting helps collect and store rainwater for future use.
- AI-Based Rainwater Harvesting Potential Predictor estimates collectible rainwater using rainfall, rooftop area, and runoff coefficient.
- This report reviews 12 research papers on rainwater harvesting, GIS, and Machine Learning techniques such as Random Forest, Gradient Boosting, and XGBoost.
- It explains the system's working, techniques, advantages, limitations, and project initiation plan to support efficient water conservation.

1. Introduction

- Water is one of the most essential natural resources required for human life, agriculture, and environmental sustainability. Increasing population, urbanization, and irregular rainfall have made water scarcity a major concern.
- Rainwater harvesting is an effective method of collecting and storing rainwater for future use. Traditional methods estimate harvesting potential using rainfall, rooftop area, and runoff coefficient.
- Artificial Intelligence (AI) and Machine Learning (ML) can help analyse rainfall data and predict the quantity of rainwater that can be collected from a particular area.
- The proposed AI-Based Rainwater Harvesting Potential Predictor aims to estimate collectible rainwater and help households, farmers, and institutions plan suitable storage systems. This project promotes water conservation and efficient water resource management.

2. Literature Survey

12 research papers (2017–2026) were reviewed:

- **GIS and ML:** Choke Watershed (2026); Upper Tekeze Basin (2026) – ML-based harvesting zone identification.
- **GIS / AHP / ML:** Jaipur (2026); Bangladesh (2025) – suitable site identification.
- **Rainwater Estimation:** Shrigonda (2024); South Indian University (2021) – rooftop harvesting potential.
- **AI-Based Water Management:** Purulia (2024); Pune (2022) – water scarcity and groundwater recharge.
- **GIS and Rooftop Assessment:** Mouli et al. (2020); Pandya et al. (2018) – rooftop analysis and harvesting planning.
- **Sustainable Water Management:** Semi-Arid Tropics (2022); Singh and Turkiya (2017) – sustainable and rural water supply.

Detailed comparison is shown in Table 2.1 (next two slides).

Table 2.1: Literature Review (1/2)

No.	Paper (Author, Year)	Approach Type	Technologies Used	Key Objective
1	Choke Watershed Study (2026)	Ensemble ML	RF, GB, XGBoost	Identify suitable harvesting zones
2	Upper Tekeze Basin Study (2026)	Geo-hydrological ML	GIS, RF	Map potential harvesting zone
3	Jaipur Study (2026)	Integrated GIS-ML	GIS, AHP, ML	Identify suitable harvesting sites
4	Bangladesh Study (2025)	Comparative ML	RF, BRT, KNN, NB, AHP	Compare zone identification methods
5	Shrigonda Study (2024)	Potential Estimation	Rainfall, rooftop area	Estimate rooftop rainwater potential
6	Purulia Study (2024)	AI and GIS	ANN, RF, GIS	Assess harvesting suitability

Table 2.1: Literature Review (2/2)

No.	Paper (Author, Year)	Approach Type	Technologies Used	Key Objective
7	Mouli et al. (2020)	Rooftop Assessment	Satellite imagery, GIS	Estimate rooftop harvesting potential
8	South Indian University Study (2021)	Feasibility Analysis	Rainfall, roof area	Estimate annual collection potential
9	Vaddadi et al. (2022)	Groundwater Recharge	Rooftop harvesting analysis	Assess groundwater recharge potential
10	Semi-Arid Tropics Study (2022)	Sustainable Planning	Rainfall-runoff analysis	Identify harvesting interventions
11	Singh and Turkiya (2017)	Rural Water Supply	Rooftop harvesting assessment	Evaluate domestic water potential
12	Pandya et al. (2018)	GIS-Based Planning	Google Earth, QGIS	Plan rooftop harvesting structures

3. Information to seminar topic

- **Overview:** AI-based prediction of rainwater harvesting potential.
- **Rainwater Factors:** Rainfall, rooftop area, runoff coefficient, soil type, and slope.
- **Need:** Water conservation and reduction of water scarcity.
- **Working Principle:** Input Data → Pre-processing → ML Model → Prediction.
- **Architecture:** Data Collection, Pre-processing, Prediction Model, and Result Display.
- **Techniques:** Random Forest, Gradient Boosting, XGBoost, ANN, and GIS.
- **Evaluation Metrics:** Accuracy, precision, recall, and F1-score.
- **Challenges:** Limited data, rainfall variation, and prediction errors.

4. Advantages / Disadvantages

Advantages

- Helps conserve water and reduce water scarcity.
- Estimates rainwater collection potential.
- Supports better harvesting system planning.
- Saves time through automatic prediction.
- Helps identify suitable harvesting zones.
- Supports sustainable water management.

Disadvantages

- Requires accurate rainfall and rooftop data.
- Prediction may be inaccurate.
- Weather changes affect results.
- GIS data may not always be available.
- ML models require sufficient data.
- Results depend on the selected model and input data.

5. Conclusion

- AI-Based Rainwater Harvesting Potential Predictor is a useful approach for estimating rainwater collection potential using rainfall, rooftop area, and environmental data.
- Machine Learning techniques such as Random Forest, Gradient Boosting, and XGBoost can help improve prediction, while GIS supports the identification of suitable rainwater harvesting zones.
- This system can help households, colleges, farms, and urban areas plan rainwater harvesting more effectively. It supports water conservation, reduces water wastage, and helps address water scarcity.
- However, prediction accuracy depends on the quality of input data and changing weather conditions. Overall, this approach promotes sustainable water management and efficient use of rainwater resources.

6. Project Initiation

- **Project:** AI-Based Rainwater Harvesting Potential Predictor
- **Aim:** A system that uses rainfall, rooftop area, and environmental data to predict rainwater harvesting potential.

This section covers:

- 6.1 Description of Problem Statement
- 6.2 Scope and Objectives
- 6.3 Stakeholders
- 6.4 Seminar Discussion Plan
- 6.5 Platform / Equipment / Resources

6. 1 Description of problem statement

- Rainwater is often wasted due to a lack of proper collection and storage systems.
- It is difficult to estimate rainwater harvesting potential accurately.
- Traditional methods may require manual calculations and environmental data.
- Therefore, an AI-based system is needed to predict rainwater harvesting potential using rainfall, rooftop area, and other factors.
- This system can help users plan efficient rainwater collection and reduce water scarcity.

6. 2 Scope and objectives.

Scope

- Develop an AI-based system to predict rainwater harvesting potential.
- Estimate rainwater collection using rainfall and rooftop area.
- Consider environmental factors such as soil type and slope.
- Display the predicted rainwater collection potential.
- Support homes, colleges, farms, and urban areas.

Objectives

- To study existing rainwater harvesting methods.
- To collect rainfall and rooftop area data.
- To apply Machine Learning techniques for prediction.
- To identify suitable rainwater harvesting locations using GIS.
- To support water conservation and efficient rainwater management.

6. 3 Stake holder

- **Homeowners:** Estimate rainwater collection potential for their rooftops.
- **Farmers:** Plan rainwater harvesting for agricultural use.
- **Engineers and Planners:** Identify suitable harvesting locations.
- **Government Departments:** Support water conservation planning.
- **Researchers:** Study AI-based rainwater prediction methods.
- **Project Guide:** Guide the project and evaluate its progress.

6. 4 seminar exsicussion plan

- Introduction to AI-Based Rainwater Harvesting Potential Prediction.
- Need for AI in rainwater harvesting.
- Literature review of 12 research papers and comparison.
- Working principle and system architecture.
- Techniques, technologies, and evaluation metrics.
- Advantages, disadvantages, and challenges.
- Conclusion and project initiation.
- Question and answer session.

6.5 Platform/ Equipment/ Resources

Platform

- Windows or Linux operating system.
- Python programming language.
- Visual Studio Code for development.
- Google Colab for ML model training, if required.

Equipment

- Computer or laptop with a modern processor.
- Minimum 8 GB RAM recommended.
- Internet connection for datasets and libraries.
- Sufficient storage for data and model files.

Table 6.1: Resources

Resources	Name	Purpose
Operating System	Windows 11	Run development tools and software.
Programming Language	Python 3.11	Develop prediction models and application code.
Code Editor	Visual Studio Code	Write, debug, and test code.
ML Library	Scikit-learn	Build and evaluate ML models.
ML Algorithm	Random Forest, XGBoost	Predict rainwater harvesting potential.
Data Handling	Pandas, NumPy	Clean and process rainfall data.
Data Visualization	Matplotlib	Display data and prediction results.
GIS Tools	QGIS	Identify suitable harvesting zones.
Dataset	Rainfall and rooftop area data	Train and test prediction models.

Bibliography

Books

1. Rainwater Harvesting: In Urban Centers within the Hard Rock Terrain of the Deccan Basalts of India — Anil Lalwani, Springer, 2022.
2. Rainwater Harvesting—Building a Water Smart City — Aysha Akter, Springer, 2022.
3. Harvesting Rainwater from Buildings — Syed Azizul Haq, Springer, 2017.
4. Managed Groundwater Recharge and Rainwater Harvesting: Outlook from Developing Countries — Edited by Dipankar Saha, Karen G. Villholth, and Mohamed Shamrukh, Springer, 2024.

Websites

1. Google Scholar – Research papers on AI-based rainwater harvesting prediction and rainfall analysis.
2. ResearchGate – Research articles on rainwater harvesting, water conservation, and machine learning.
3. IEEE Xplore – Technical papers on Artificial Intelligence, Machine Learning, and predictive modelling.
4. ScienceDirect – Research articles on rainwater harvesting, GIS techniques, and machine learning-based prediction.

References

1. “Rainwater Harvesting Potential Zone Identification Using GIS – Machine Learning Approach and Comparison of Their Performance,” 2026.
2. “Integrating Machine Learning and Geo-Hydrological Modeling to Identify Rainwater Harvesting Potential Zones in the Upper Tekeze Basin,” 2026.
3. “Integrated Machine Learning and AHP-Based Framework for Spatial Delineation of Rainwater Harvesting Potential,” 2026.
4. “Assessing the Performance of Machine Learning and AHP Models for Rainwater Harvesting Potential Zone Identification in Hilly Region, Bangladesh,” 2025.
5. “Estimation of Roof Top Rain Water Harvesting Potential: A Proposed Solution to Water Scarcity in Shrigonda Town, Maharashtra, India,” 2024.
6. “Addressing Water Scarcity Challenges Through Rainwater Harvesting: A Case Study on Purulia, India,” 2024.
7. T. Mouli, K. Krupavathi and Ch. S. Ganeswara Rao, “Urban Rooftop Area Assessment for Estimation of Rooftop Rainwater-Harvesting Potential,” 2020.
8. “Feasibility of Roof Top Rainwater Harvesting Potential – A Case Study of South Indian University,” 2021.
9. N. Vaddadi, C. Vansarochana and V. Raghavan, “Estimation of Groundwater Recharge Potential Using Rooftop Rainwater Harvesting: Case Study from Pune Urban Area, India,” 2022.
10. “Identifying Potential Zones for Rainwater Harvesting Interventions for Sustainable Intensification in the Semi-Arid Tropics,” 2022.
11. Omvir Singh and Sushila Turkiya, “Assessing Potential for Rooftop Rainwater Harvesting: An Option for Sustainable Rural Domestic Water Supply in Arid Region of Haryana,” 2017.
12. P. A. Pandya et al., “Planning of Rooftop Rainwater Harvesting Structure Based on GIS Approach,” 2018.

Thank you